

CS 411 - Artificial Intelligence I
Fall 2025
Assignment 3
Department of Computer Science, University of Illinois at Chicago

Total Points: 25

15 puzzle is a sliding puzzle game with numbered squares arranged in 4X4 grid with one tile missing.

Initial State

1		2	4
5	7	3	8
9	6	11	12
13	10	14	15

The puzzle is solved when the numbers are arranged in order.

Goal State

1	2	3	4
5	6	7	8
9	10	11	12
13	14	15	

The actions are defined in terms of direction where empty square can be moved to

UP (U), Down(D), Left(L), Right(R)

Write a program which performs a breadth-first search to find the solution to any given board position for 15 puzzle

Input

The input should be given in the form of a sequence of numbered tiles for initial board configuration, '0' indicating the empty space (see example below)

Output

1. Moves
2. Number of Nodes expanded
3. Time Taken
4. Memory Used

Example

> 1 0 2 4 5 7 3 8 9 6 11 12 13 10 14 15

Moves: RDLDDRR

Number of Nodes expanded: 361

Time Taken: 0.238

Memory Used: 704kb

Hint

- You can use hashset to keep track of explored nodes and fast lookup

Submission

1. Please submit a zip file named bfs.zip, containing the following files:

- Source Code
- Readme.txt with instructions on how to run the code

2. Zipping Instructions:

- Do not place the files inside a folder before zipping.
- Select both files directly and zip them together.

Submission Platforms:

You must submit your assignment on both Gradescope and Blackboard.

Programming Language

You can choose from C++, Java, Python or Julia

Rubric

Print the moves to reach the solution => 10

Check for repeated states => 3

Print number of nodes expanded => 3

Print total memory usage => 3

Print total time taken => 3

Coding style, comments, readme instruction => 3

Your submission should be your own work. You can take reference from aima codebase but you are not allowed to use the code directly. All submissions will be checked for plagiarism using MOSS (Measure Of Software Similarity).